MINOR IN INDUSTRIAL ENGINEERING AND MANAGEMENT SYSTEMS

Rationale

The Industrial Engineering and Management Systems (IEMS) minor aims to expand engineering students' expertise in system-wide optimization, decision science, and digital integration. It prepares students to architect the "intelligence layer" of the enterprise and prepares undergraduate students for graduate studies in Industrial Engineering, Engineering Management, and Operations Research.

It primarily targets mechanical and mechatronics engineering students, but it also accommodates students from various engineering backgrounds, provided all prerequisite courses are completed. The program focuses on mathematical optimization, financial engineering, facility logistics, quality assurance, and digital twin simulation. By the end of this minor, a student is expected to be proficient in designing, simulating, and managing complex industrial ecosystems that are cost-effective, data-driven, and industry-ready.

Program participants are required to complete at least 18 credits of related coursework. Although not required, with proper and timely advising, they may also complete their Co-op work experience or their BE project in Industrial Engineering or Systems Management, thereby enhancing their capabilities.

Program Objectives

The primary educational objectives of the IEMS minor program are to:

- Give engineering students a sophisticated understanding of Industrial Engineering principles and management systems.
- Equip students with the knowledge to pursue graduate studies in Industrial Engineering, Engineering Management, Systems Engineering, Data Analytics, and other related disciplines.

Learning Outcomes

After completing the minor program, students should have the ability to:

- Apply advanced mathematical modeling and operations research to optimize resource allocation and industrial scheduling.
- Analyze the financial viability of engineering projects using rigorous economic tools and ROI assessments.

- Design and validate complex systems using CAD-integrated Digital Twin and discrete-event simulation software.
- Manage global supply chains and logistics networks while ensuring high standards of quality through Six Sigma and Lean methodologies.

Career Options

This minor allows IEMS holders to start careers in engineering management, systems analytics, supply chain architecture, industrial consultancy, project management, operations strategy, and digital manufacturing integration in both regional and global markets.

Program Requirements

While most relevant to mechanical and mechatronics engineering students, this minor is offered to undergraduate RHU students in the BS/BE program from different disciplines. It may also be offered to students with a BS/BE from other universities, subject to a case-by-case evaluation of their transcripts. Proper advising is pertinent to the timely completion of the minor. To complete the IEMS minor, a student must:

1. Declare an IEMS minor by completing the Minor Declaration Form;
2. Obtain the approval of the dean of the college major;
3. Obtain a Minor Cumulative Grade Point Average of no less than 70%;
4. Complete 18 credits of coursework as specified below.

IEMS Curriculum

The program offers a rich blend of mathematical optimization and virtual system modeling, equipping students with the tools to navigate the modern industrial world. Accordingly, this minor shall include:

- Engineering financial management and project economics.
- Advanced quality engineering and lean operations.
- Mathematical optimization and deterministic operations research.
- Physical facility planning and systematic layout design.
- Digital Twin architecture and virtual system simulation.
- Global supply chain engineering and logistics optimization.

Mandatory Courses

The student must complete a minimum of 6 credit hours of coursework exclusive to the IEMS minor that will not count toward requirements for their major or any other minor. A description of the 18 credits from which a student must select is given below:

ENGR 300	Engineering Economics and Management	3(3,0)
Concepts and techniques in basic Engineering economy principles and applications. Interest and financial mathematics: present worth, annual worth, benefit/cost ratio, internal rate of return, multiple alternatives, income tax, inflation, Risk analysis, Investment and investment choice, equivalence, loans, cost of capital, retirement and replacement, sector analysis and viewpoint, sensitivity analysis, accounting, and financial statements.		

BMGT 410	Quality Management	3(3,0)
This course highlights that TQM has become crucial for business excellence in worldwide markets. It allows students to learn techniques that establish sustainable quality improvement in both product and service industries. Topics include seven quality control tools; SPC; quality function deployment and the house of quality; product reliability; Six Sigma; Benchmarking; quality certification and awards; and quality costing. **Prerequisite:** BADM 250		

ENGR 400	Operations Research	3(3,0)
This course introduces mathematical modeling and algorithmic techniques for optimizing decision-making in complex engineering systems. Students will master the application of Linear Programming and the Simplex Method to solve problems involving constrained resources. Key topics include duality and sensitivity analysis (determining how changes in the system affect the optimal solution), network flow optimization (shortest-path and PERT/CPM), and integer programming. The course emphasizes translating real-world industrial challenges—such as production scheduling, trajectory planning for robots, and resource allocation—into rigorous mathematical frameworks. **Prerequisite:** MATH 311		

ENGR 410	Supply Chain Engineering & Logistics	3(3,0)
This course applies engineering and mathematical principles to the design and optimization of global supply chain networks. Moving beyond general procurement, students will focus on the quantitative modeling of inventory control (including deterministic and stochastic models), warehousing operations, and transportation logistics. Key topics include the "Bullwhip Effect" in multi-echelon systems, facility location modeling using center-of-gravity methods, and the strategic integration of Just-in-Time (JIT) principles. The course specifically addresses Design for Logistics (DFL), teaching students how mechanical design decisions (packaging, weight, and modularity) impact the total landed cost and supply chain resilience in a global market. **Prerequisite:** MATH 351 and ENGR 300		

MECH 424	Manufacturing Systems & Facilities Planning	3(3,0)
This course focuses on the strategic design and organization of integrated manufacturing environments. Moving from individual machine operations to system-wide flow, students will study Systematic Layout Planning (SLP), group technology, and the design of Cellular Manufacturing units. The curriculum emphasizes quantitative methods for line balancing, machine-requirement calculations, and material-handling systems. Students will analyze the physical constraints of the shop floor to minimize "waste" in movement and optimize the integration of automated workstations and robotic cells. **Prerequisite:** MECH 224 and MECH 422		

MECA 425	Digital Twin & System Simulation	3(3,0)
This course explores the creation and use of Digital Twins to model, simulate, and predict the behavior of complex manufacturing and mechatronic systems. Utilizing discrete-event simulation (DES) software, students will develop virtual replicas of physical systems to analyze the impact of stochastic variables, such as machine downtimes and fluctuating arrival rates. The course covers data distribution fitting, model verification, and the statistical validation of simulation outputs. By performing "what-if" experimental analyses, students will learn to identify system bottlenecks and optimize throughput in a risk-free virtual environment before physical deployment. **Prerequisite:** MATH 351 and MECA 541		